

MathVR: Teaching Vector Arithmetic Using Virtual Reality

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Figure 1: A full tutorial class engaging with the MathVR activity.

ABSTRACT

Traditionally, linear algebra has been taught using 2D representations of fundamentally 3D concepts. We present a visualisation of vector arithmetic in virtual reality with full hand presence and lateral motion, demonstrating how this new technology can be used to enhance student engagement and understanding. We detail sufficiently powerful affordances to convey ideas such as compound operations, yet simple enough to be digestible by average undergraduate students in a 1st year university mathematics course. This presentation aims to catalyse discussion and ideas around the use of immersive technologies for education in more abstract disciplines such as mathematics.

CCS CONCEPTS

• **Human-centered computing** → **Visualization; Interaction design.**

KEYWORDS

virtual reality, mathematics education, linear algebra, vector arithmetic

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1 INTRODUCTION

The recent resurgence of virtual reality (VR) technology has highlighted its immense potential for applications in education and training. Many studies have already demonstrated the efficacy of VR training for task oriented training that would have concrete outcomes in the real world, such as in mining [Zhang 2017], surgery [Bric et al. 2016], construction [Wang et al. 2018] and logistics [Yigitbas et al. 2020]. However, VR's potential for illuminating more abstract concepts is not as well recognised. VR presents tantalising prospects for new and effective ways of representing ideas, not previously possible.

Mathematics is one such prospect [Abraham Al-bashki 2003; Kaufmann et al. 2000; Su et al. 2022]. This potential is particularly apparent in linear algebra, especially for 3D vector arithmetic. Traditionally, this topic is taught using 2D representations on paper or the blackboard. In VR, this topic can be represented in its native 3D form, eliminating any semantic distance that is necessitated by the use of an incomplete analogy (i.e. 2D diagrams) due to the limitations of the medium employed (i.e. paper, blackboard).

We present MathVR, a standalone VR application used to aid in the teaching of vector arithmetic. In 2020, prior to the first COVID-19 pandemic lockdowns in March, MathVR was used as an assessed tutorial activity in MATH1131 and MATH1141 at the University of New South Wales. MathVR was also presented at the Games 4 Change APAC Festival (online) in 2021. It takes advantage of the possibilities provided by 6-DoF hand presence for manipulating vectors and representing fundamental operations.

2 AFFORDANCES

A single vector is represented by an arrow in space, anchored by a box. Moving the box moves the representation's location. Moving the arrow head modifies the vector's value within the simulation. The manipulations are intuitively made using natural grab hand gestures. (See Figure 2.) As many vectors as required can be generated from a floating virtual control panel.

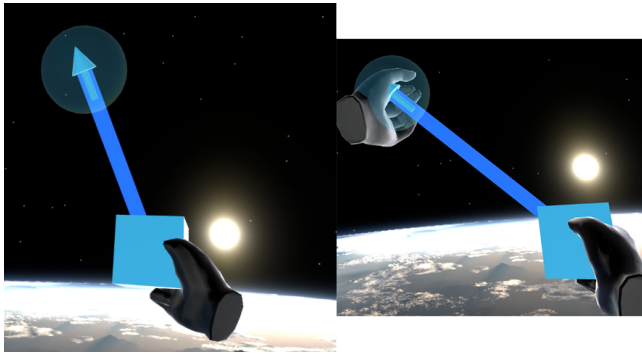


Figure 2: Representation of a vector in MathVR

2.1 Operations

When two vectors are placed within an operation object, the results are represented by another non-editable vector emanating from the centre of the operation object. The vectors being used as parameters can still be manipulated, and the operation's result is updated in real-time. (See Figure 3).

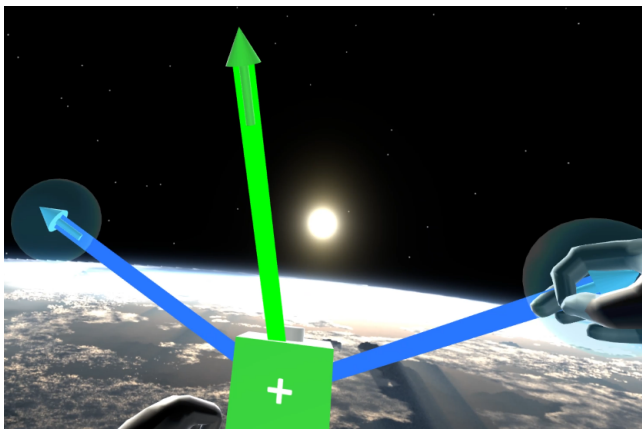


Figure 3: An addition operation, defined by two vectors.

The operations available are:

- Addition
- Subtraction
- Projection
- Cross Product

2.2 Cascading Results

Operation results can be cloned and exist as independently movable, but non-editable vectors. This can be useful for representing key geometric characteristics. For example, the geometric significance of the subtraction operation can be made more apparent by its result being placed at the head of the first parameter (See Figure 4.)

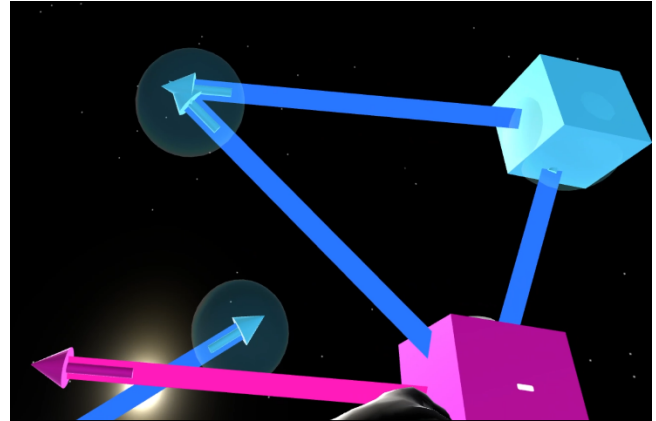


Figure 4: The subtraction operation demonstrates how its result is the direction from one vector to another.

Cloned results can be used as parameters in subsequent operations, which allows for chaining the results. In Figure 6, the result of a projection operation is placed into an addition operation, which is in turn placed into a subtraction operation. This sequence can continue to proceed in perpetuity (within hardware limits).

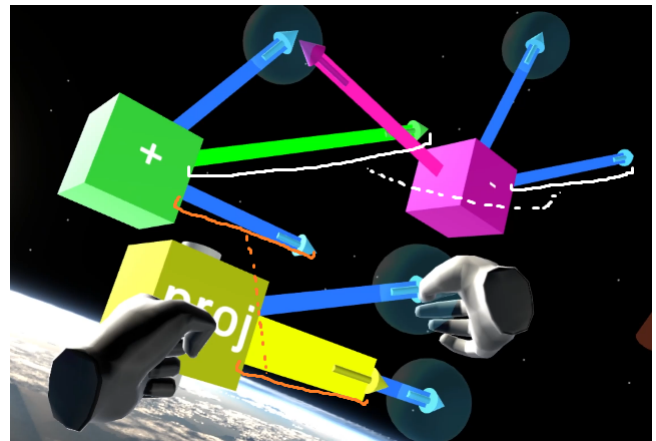


Figure 5: Note the coloured annotations tracing the sequence of operations.

3 FUTURE WORK

There has been positive informal verbal feedback from students that have gone through this experience. We intend to more robustly evaluate the effect and efficacy of MathVR to verify this feedback. As we emerge from the pandemic, on-campus activities such as

MathVR are once again feasible. A phone based AR version may also be developed to enhance its accessibility for students.

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A LIST OF SPECIFIC TASKS IN ACTIVITY

- (1) add two vectors
- (2) subtract two vectors
- (3) delete a vector
- (4) spawn a plane
- (5) copy plane normal
- (6) project onto normal
- (7) move tail of normal projection onto plane

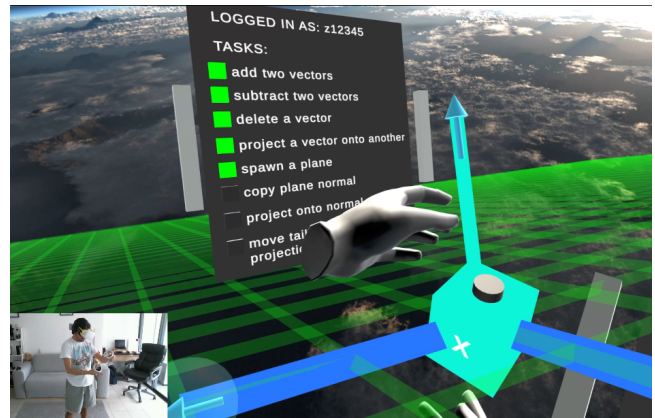


Figure 6: Students must complete specific tasks, like manipulating a vector defined plane.